

DULUTH SKY HARBOR, MN, USAWMO: **727456**Lat: **46.722N**Lon: **92.043W**Elev: **610**StdP: **14.37**Time Zone: **-6.00 (NAC)**Period: **98-19**WBAN: **04919****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-6.4	-19.8	1.9	-9.0	-16.9	2.2	-6.0	28.3	24.8	25.7	25.1	8.1	220	0.644

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.8	86.1	71.5	82.3	69.2	80.6	67.9	75.0	82.5	72.5	79.3	70.1	76.7	12.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.5	123.2	79.0	70.3	114.1	77.0	67.6	103.9	74.4	38.9	82.5	36.5	78.5	34.5	75.2	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.5	25.2	21.8	-16.8	92.0	4.5	3.7	-20.0	94.7	-22.6	96.9	-25.1	99.0	-28.4	101.7		
(5)			-16.8	78.6	4.0	3.1	-19.7	80.9	-22.1	82.7	-24.3	84.4	-27.2	86.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.4	15.9	18.5	29.1	39.2	48.6	58.2	68.4	66.9	59.3	47.1	34.9	21.3
	DBStd	19.95	12.49	11.14	10.01	7.08	7.49	7.87	6.62	5.72	7.23	7.25	9.33	11.42
	HDD50	4549	1057	882	648	332	120	15	0	0	8	140	457	890
	HDD65	8552	1522	1302	1113	773	511	230	43	45	199	555	904	1355
	CDD50	1783	0	0	0	9	77	262	569	523	288	52	3	0
	CDD65	309	0	0	0	0	2	27	148	102	29	1	0	0
	CDH74	2327	0	0	0	2	65	270	1175	618	187	10	0	0
	CDH80	651	0	0	0	0	13	65	373	157	42	1	0	0
	WSAvg	9.2	8.9	9.1	8.7	10.5	9.4	8.6	8.1	8.1	9.2	10.2	9.8	9.3
	PrecAvg	30.5	0.9	0.9	1.4	2.6	3.1	4.2	4.4	3.7	3.7	3.2	1.8	1.1
(15) Precipitation	PrecMax	39.0	2.1	2.1	3.1	8.7	5.6	6.7	10.0	7.6	7.3	7.0	4.5	2.3
	PrecMin	21.1	0.1	0.2	0.4	0.6	1.1	1.4	1.8	1.3	1.0	0.7	0.0	0.2
	PrecStd	4.5	0.6	0.5	0.7	1.8	1.1	1.5	2.0	1.7	1.7	1.6	1.3	0.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.5	45.7	55.2	70.3	81.6	87.6	91.0	89.9	84.2	73.2	61.2	45.0
		MCWB	37.0	39.9	44.5	52.5	62.6	74.3	75.5	76.9	71.5	60.3	50.7	39.6
	2%	DB	37.3	39.0	48.2	61.5	73.0	81.3	87.9	83.9	79.5	66.0	54.3	39.5
		MCWB	35.0	35.6	41.4	48.5	57.7	65.9	72.3	71.6	68.4	57.2	47.7	37.2
	5%	DB	35.8	36.3	44.6	54.6	66.3	76.5	83.9	81.0	74.5	62.6	50.4	37.4
		MCWB	33.8	34.1	39.1	45.2	55.5	63.8	70.6	68.9	64.8	54.7	45.6	35.8
	10%	DB	33.6	34.1	39.4	48.2	61.3	72.1	81.2	77.1	70.4	56.9	46.4	36.2
		MCWB	31.8	32.1	35.9	42.1	52.4	62.6	68.9	67.0	63.3	51.4	43.5	34.9
	0.4%	WB	38.6	41.0	47.4	55.0	65.4	74.4	78.4	78.4	74.6	64.2	53.0	41.0
		MCDB	41.6	45.4	52.8	64.1	76.2	83.5	86.3	84.9	81.7	69.6	58.4	43.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.7	36.7	42.3	50.3	60.3	68.8	75.2	74.8	70.4	59.1	49.3	38.3
		MCDB	37.7	38.6	47.4	60.5	70.5	75.6	83.4	81.3	77.0	64.4	53.3	39.8
	5%	WB	33.3	34.3	39.5	46.4	57.1	65.9	72.9	72.1	67.1	55.2	46.4	36.6
		MCDB	35.2	36.0	43.5	53.2	65.3	74.2	80.4	77.5	72.7	59.8	48.6	37.9
	10%	WB	31.0	31.8	37.1	42.9	53.4	63.3	70.6	69.2	64.4	53.0	44.5	34.6
		MCDB	32.6	33.8	39.6	47.8	59.4	70.0	78.2	74.5	67.9	56.3	47.1	36.1
(35) Mean Daily Temperature Range	5% DB	MDBR	14.5	15.5	13.5	11.6	13.6	14.6	15.8	14.3	14.0	11.9	11.1	11.8
		MCDBR	13.8	15.8	17.9	24.3	27.4	25.5	22.8	20.7	21.0	19.6	16.5	11.4
		MCWBR	11.8	12.8	12.1	14.0	15.1	13.7	11.0	10.7	11.7	12.0	11.6	9.7
	5% WB	MCDBR	12.7	13.9	16.1	22.6	24.7	22.7	20.3	17.6	18.4	16.4	14.6	10.4
		MCWBR	11.2	11.6	11.3	14.1	15.3	14.2	11.7	11.2	11.6	11.7	11.1	9.2
(40) Clear-Sky Solar Irradiance	taub		0.250	0.274	0.317	0.356	0.386	0.383	0.388	0.393	0.371	0.318	0.292	0.270
	taud		2.412	2.355	2.353	2.362	2.331	2.370	2.378	2.385	2.417	2.551	2.512	2.417
	Ebn at Noon		275	287	288	285	279	280	276	270	267	269	255	251
	Edh at Noon		21	28	33	36	39	37	37	35	31	23	20	19
(44) All-Sky Solar Radiation	RadAvg		456	799	1167	1457	1689	1830	1940	1624	1223	744	443	325
	RadStd		49	73	108	147	125	126	78	116	82	94	51	33

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days</
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